# Unit 03: Connecting to Internet

**CONDUCT**



## Objectives

After studying this unit, you will be able to understand

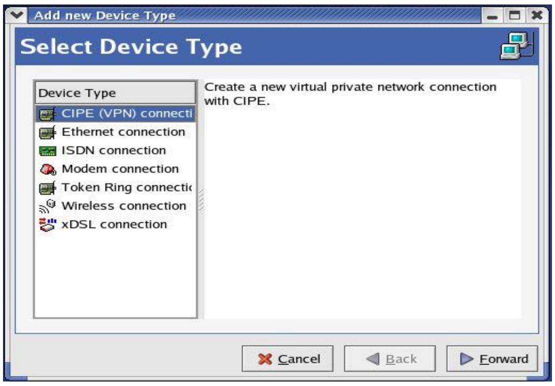
- Understand the Network Interfacing Tool
- Connect to LAN using static and dynamic addresses
- Understand DNS
- Know useful commands for configuration of system for internet
- Understand the internet connectivity in Linux

## Introduction

When the installation of operating system is completed on the computer system, the next task we do is to connect it to the internet. The surfing of websites, playing online games, sending, and receiving of emails etc. is possible only through internet. For this, first we need to configure our computer system so that the connection can take place.

## 3.1 Internet Configuration Wizard

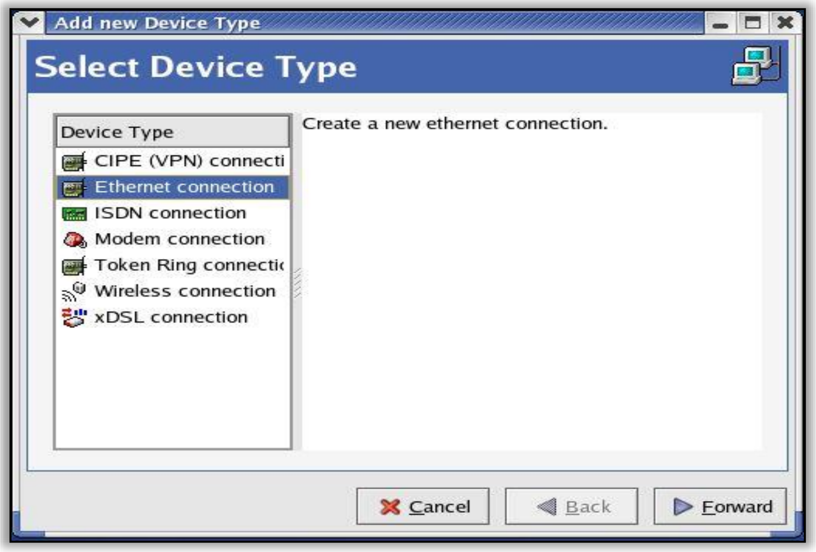
The internet configuration wizard can be opened by following this path: Main Menu | System Tools | Internet Configuration Wizard. Network interfacing tool is also known as the network administration tool or network configuration tool.



## 3.2 Connecting to LAN

A local area network (LAN) is a collection of devices connected in one physical location, such as a building, office, or home. A LAN can be small or large, ranging from a home network with one user to an enterprise network with thousands of users and devices in an office or school. A network device connected to a TCP/IP network has an IP address associated with it, such as 192.168.100.20. Using the IP address of a machine, other machines on the network can address it uniquely.A LAN comprises cables, access points, switches, routers, and other components that enable devices to connect to internal servers, web servers, and other LANs via wide area networks.

Once the network interfacing wizard is opened, then we need to select the device type. The available options are VPN connection, ethernet connection, ISDN connection, modem connection, token ring connection, wireless connection and xDSL connection. As we going to connect to LAN, so for that we need to click on 'Ethernet connection', it will create a new ethernet connection. Once selected, click on forward.



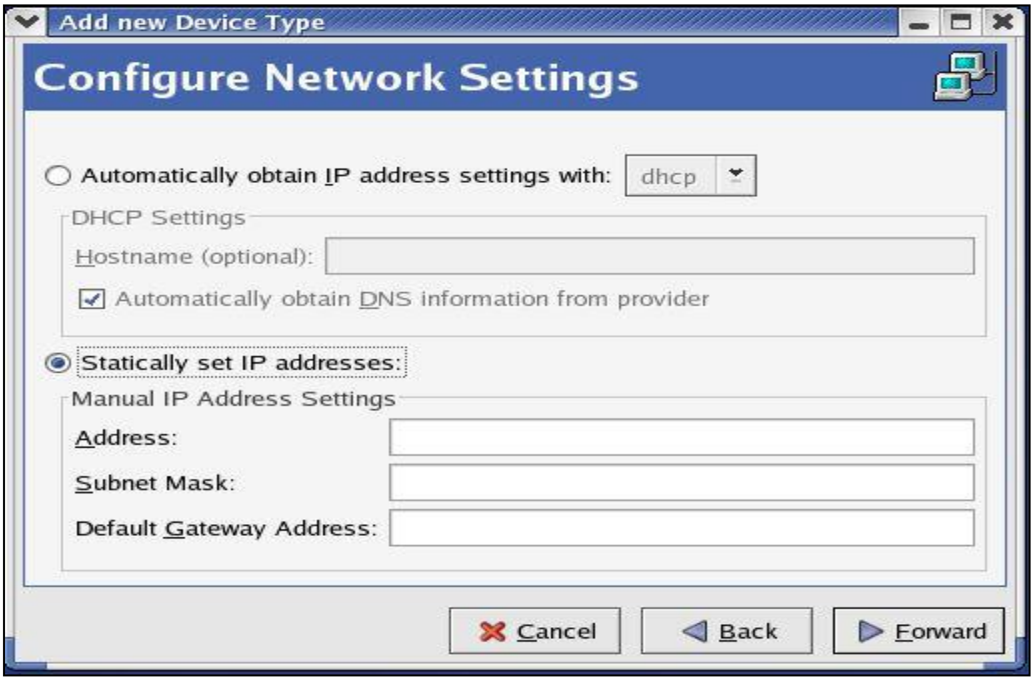

Next it will ask for the selection of Ethernet card. The options are AMD PCnet32 and other ethernet cards. Click on AMD PCnet32 (eth0) and then click forward.


Next it will ask for configuration of network settings, and these networks are provided by the means of IP addresses. An IP address is a unique address that identifies a device on the internet or a local network. IP stands for "Internet Protocol," which is the set of rules governing the format of data sent via the internet or local network. In essence, IP addresses are the identifier that allows information to be sent between devices on a network: they contain location information and make devices accessible for communication. The internet needs a way to differentiate between different computers, routers, and websites. IP addresses provide a way of doing so and form an essential part of how the internet works. These are of two types: static IP address and dynamic IP address.

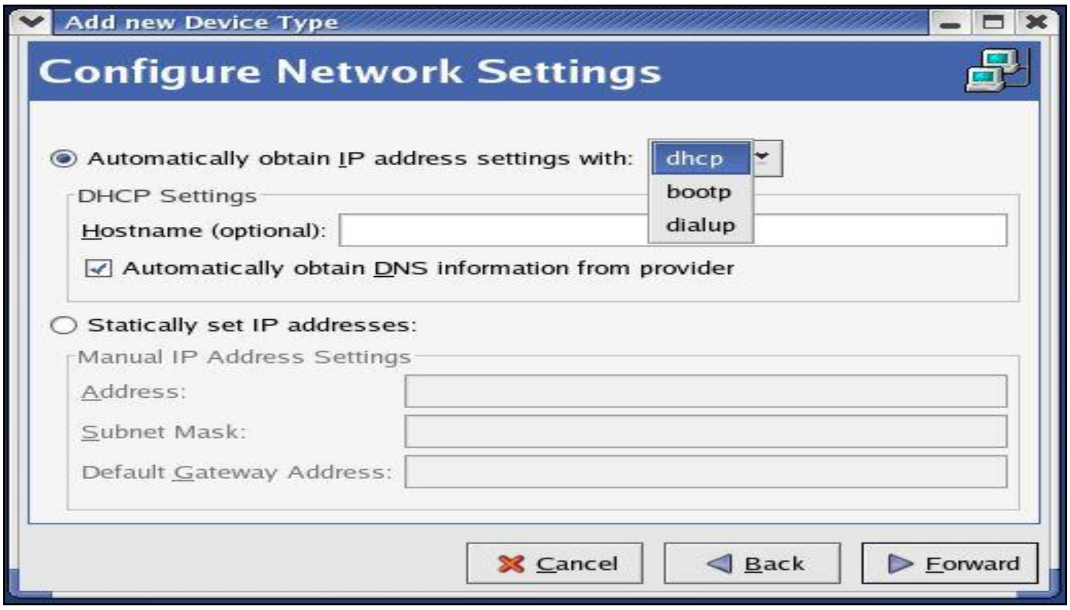
**Static IP address:** A static IP address is simply an address that doesn't change. Once your device is assigned a static IP address, that number typically stays the same until the device is decommissioned or your network architecture changes. Static IP addresses generally are used by servers or other important equipment. Static IP addresses are assigned by Internet Service Providers (ISPs). Your ISP may or may not allocate you a static IP address depending on the nature of your service agreement. We describe your options a little later, but for now assume that a static IP address adds to the cost of your ISP contract. A static IP address may be IPv4 or IPv6; in this case the important quality is static. Some day, every bit of networked gear we have might have a unique static IPv6 address. We're not there yet. For now, we usually use static IPv4 addresses for permanent addresses.

**Dynamic IP address:**As the name suggests, dynamic IP addresses are subject to change, sometimes at a moment's notice. Dynamic addresses are assigned, as needed, by Dynamic Host Configuration Protocol (DHCP) servers. We use dynamic addresses because IPv4 doesn't provide enough static IP addresses to go around. So, for example, a hotel probably has a static IP address, but each individual device within its rooms would have a dynamic IP address. On the internet, your home or office may be assigned a dynamic IP address by your ISP's DHCP server. Within your home or business network, the dynamic IP address for your devices -- whether they are personal computers, Smartphone, streaming media devices, tablet, what have you -- are probably assigned by your network router. Dynamic IP is the standard used by and for consumer equipment.

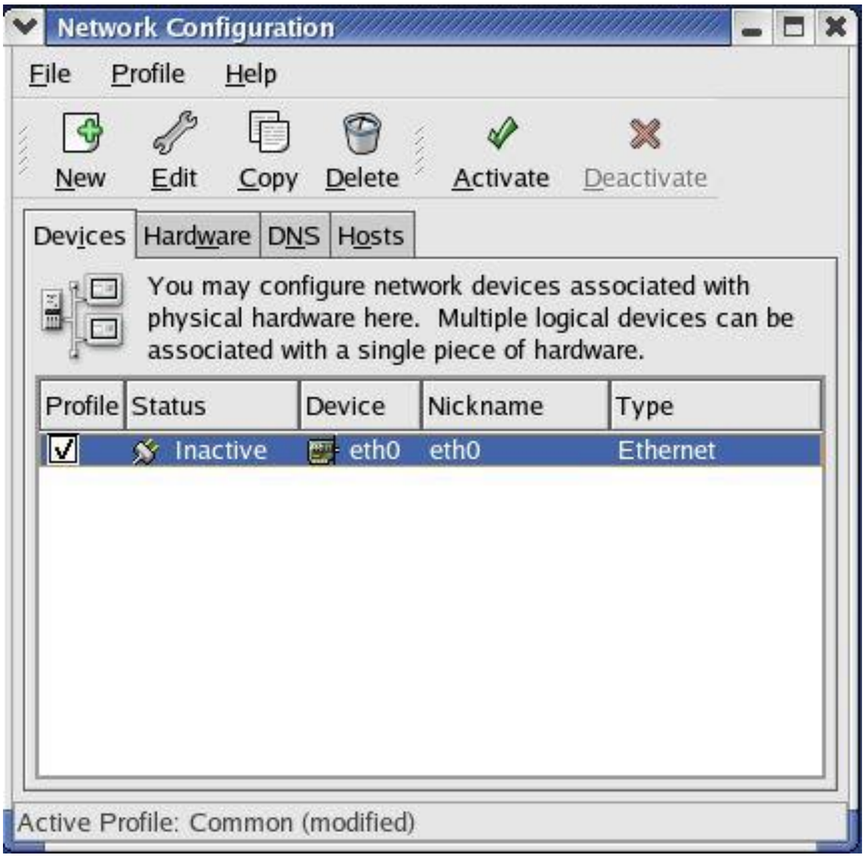
The next step is to select static or dynamic IP address. So when you go further, the next screen will show this.




Here we need to select the dynamic IP address. Dynamic IP addresses are distributed and managed via dynamic address allocation protocols. In the case of a machine connected to a LAN using a dynamic IP address, the address is allocated either using the DHCP or BOOTP. For ISPs that use PPPoE, the address is allotted by the PPPoE protocol, in which case we need to choose dialup.




When everything is done, it will create an Ethernet device.

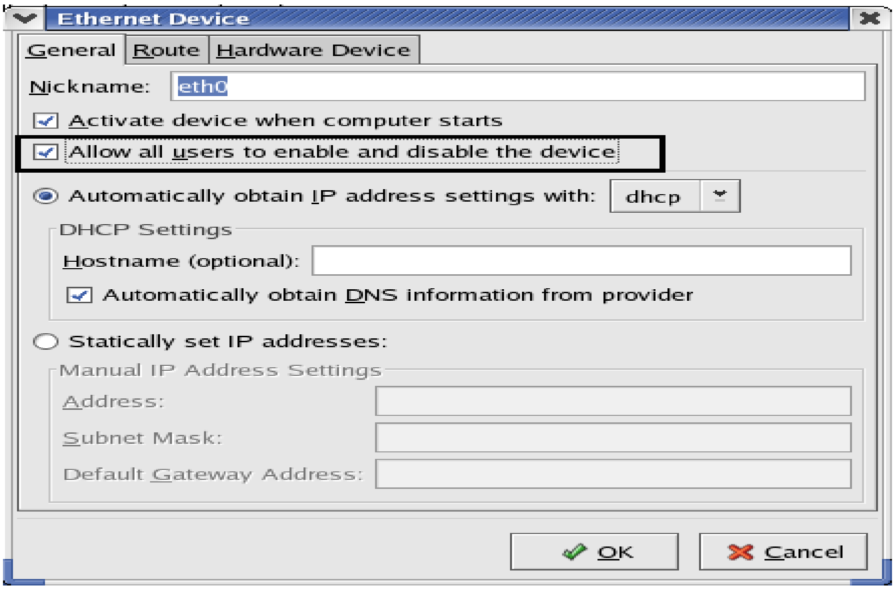



The network interfacing tool will be opened. It has four tabs:

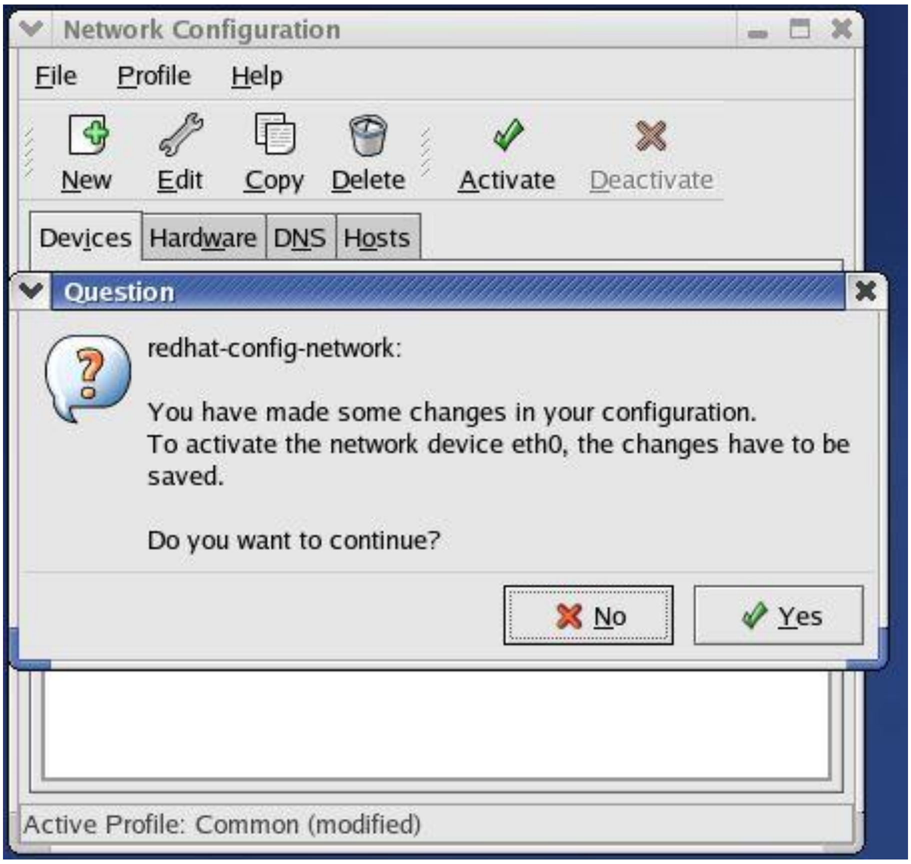
- Devices Tab: Lists the device connections that we have available on our machine.
- Hardware Tab: Allows us to manage the various network devices on the system, such as Ethernet cards, internal modems, and wireless cards.
- DNS Tab: Allows us to specify DNS server information.

- Hosts Tab: Allows us to modify the hostname of the machine and add aliases to the same host.

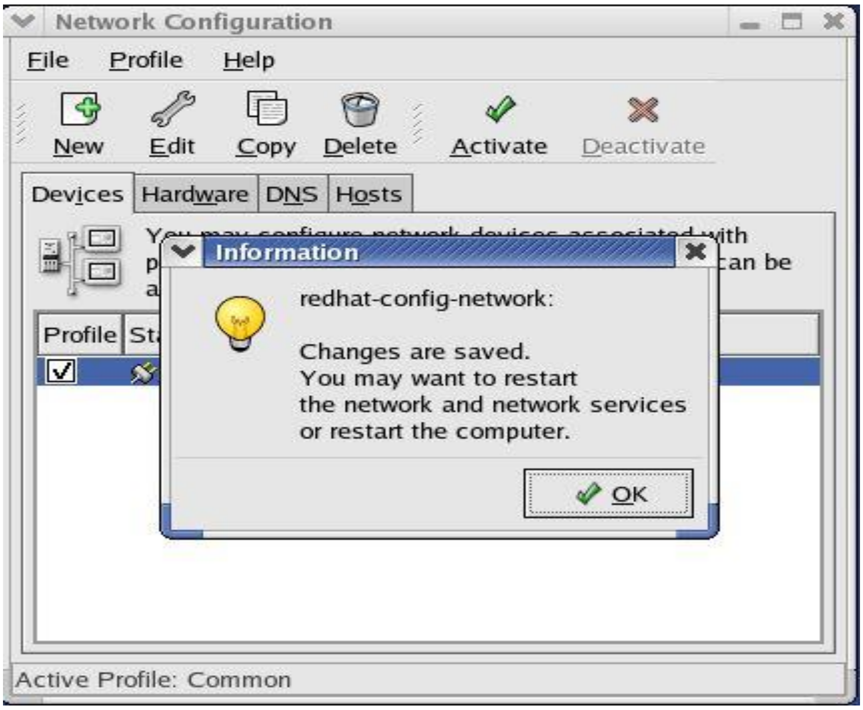
When you double click on the selected device, the next screen will look like this.



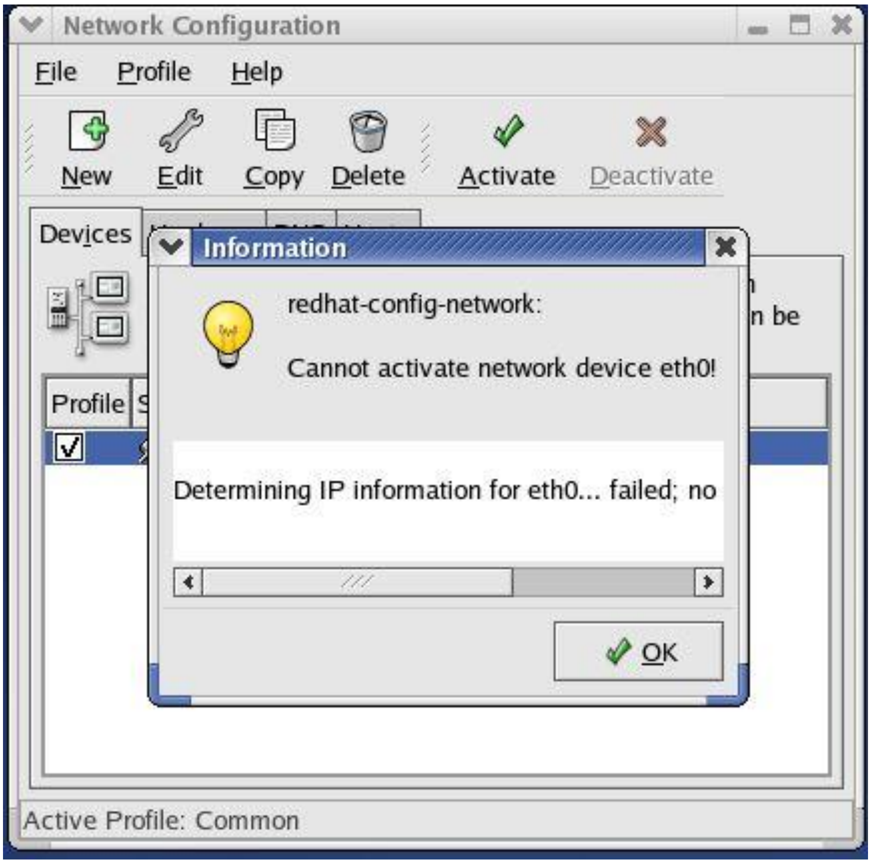

Click on allow all users to enable and disable the device. The next screen is:



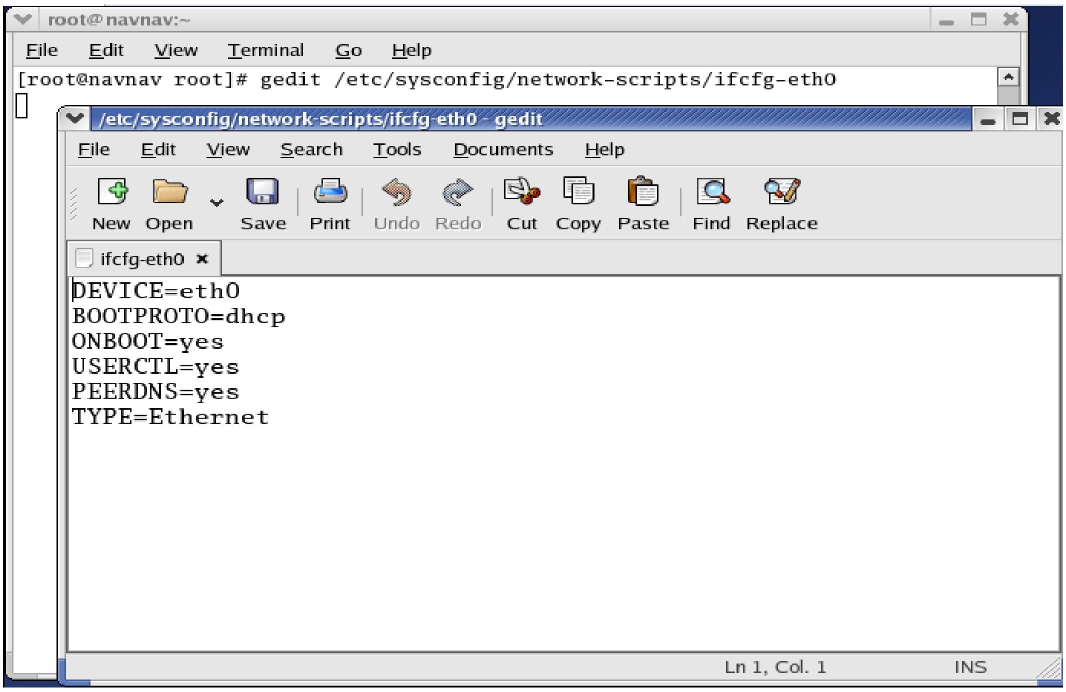

Click on OK.




To activate the internet connection, click on ok.




When this is done, our next task is to modify three files so that the computer system can be connected to the internet. These files are opened using the editors. For the opening of first file, we need to write #gedit /etc/syscon fig/network-scripts/ifcfg-eth0 in terminal and then press ENTER. When this file is opened, it will show




Here at the bottom of it, we need to add some content, i.e.,

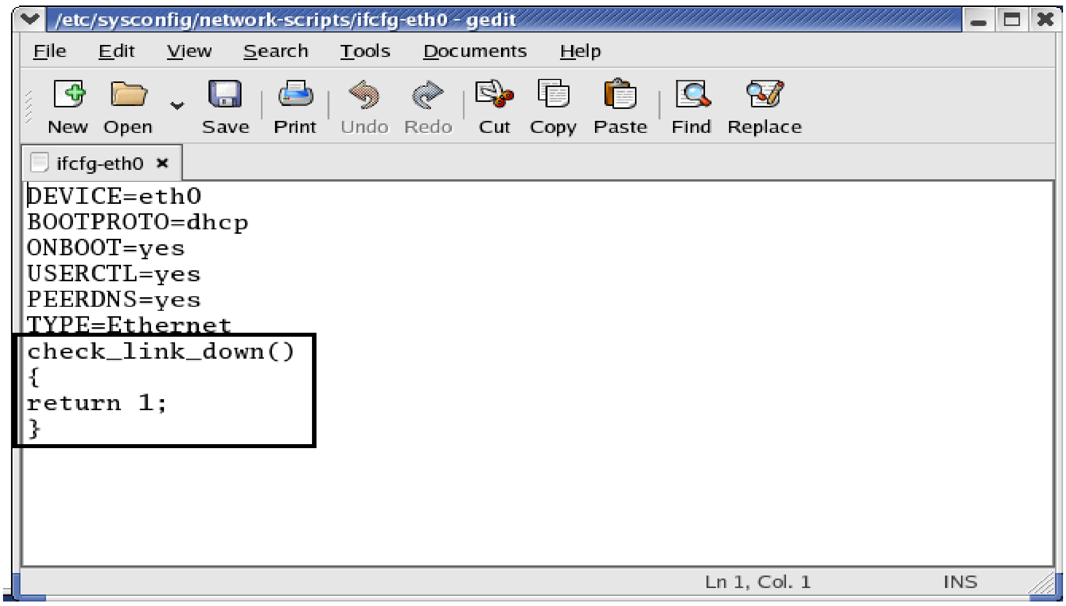
Check_link_down()

{

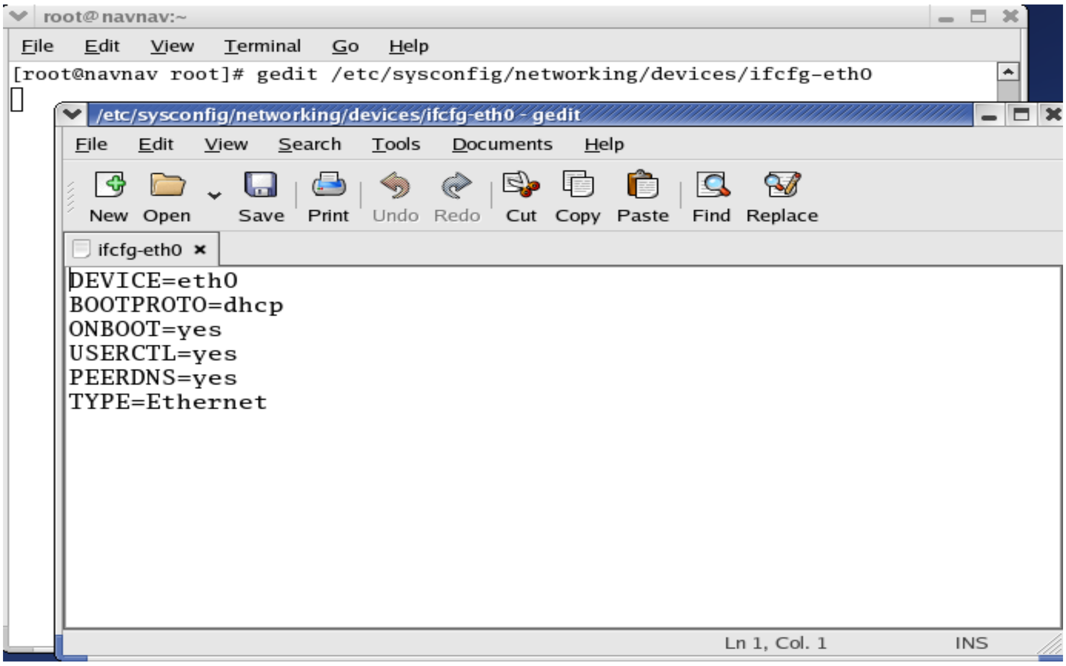
return 1;

}

So, after modification, it will look like




Next we need to open and modify second file: #gedit /etc/sysconfig/networking/devices/ifcfg-eth0.




Here at the bottom of it, we need to add some content, i.e.,

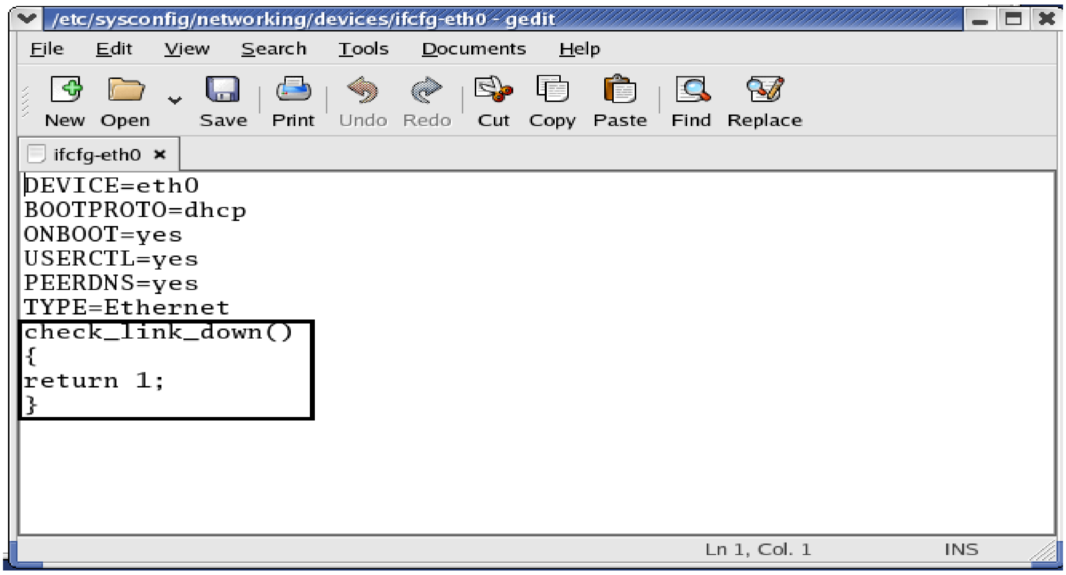
Check_link_down()

{

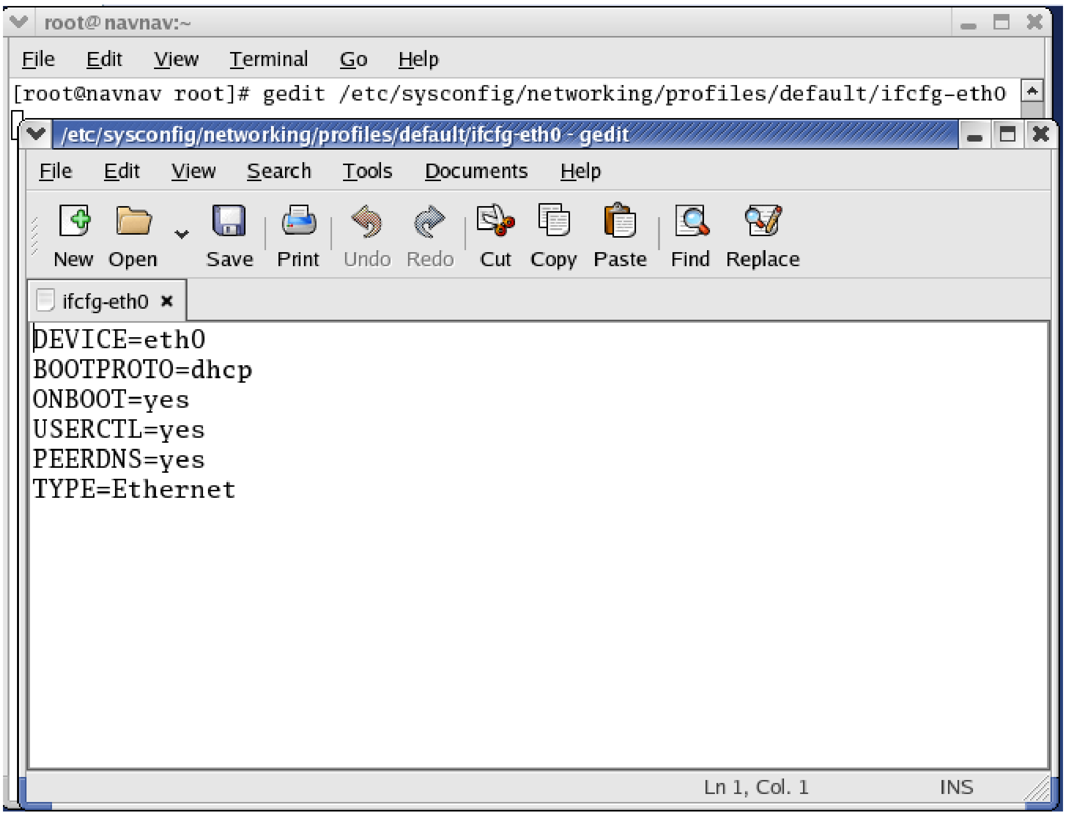
return 1;

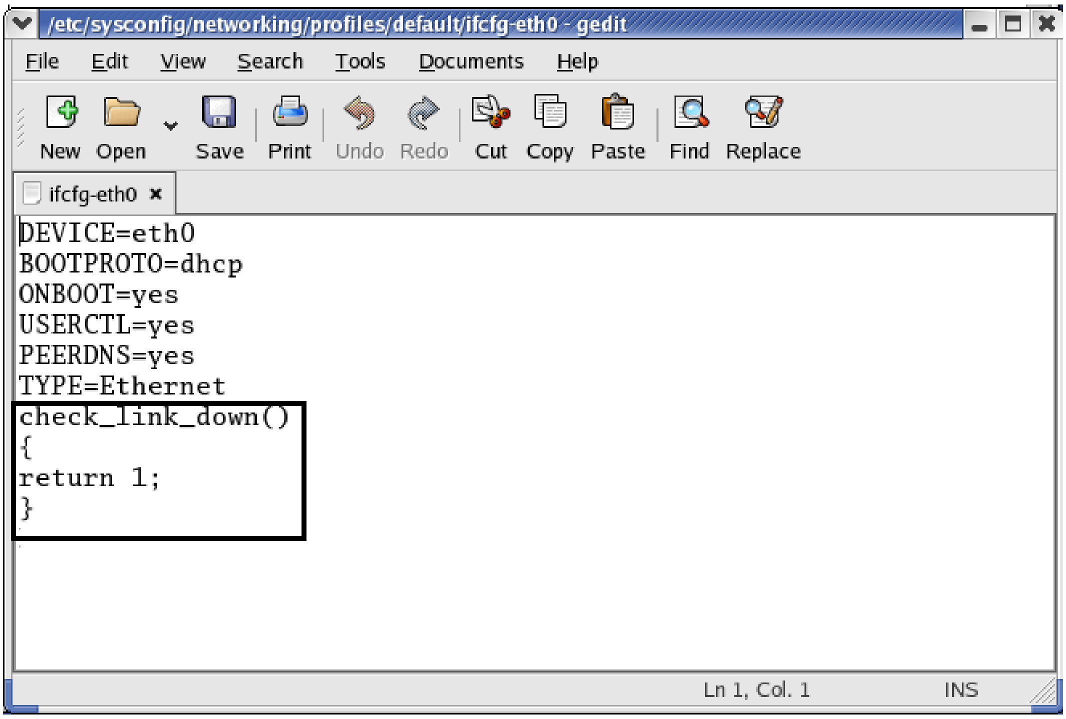
}

So, after modification, it will look like

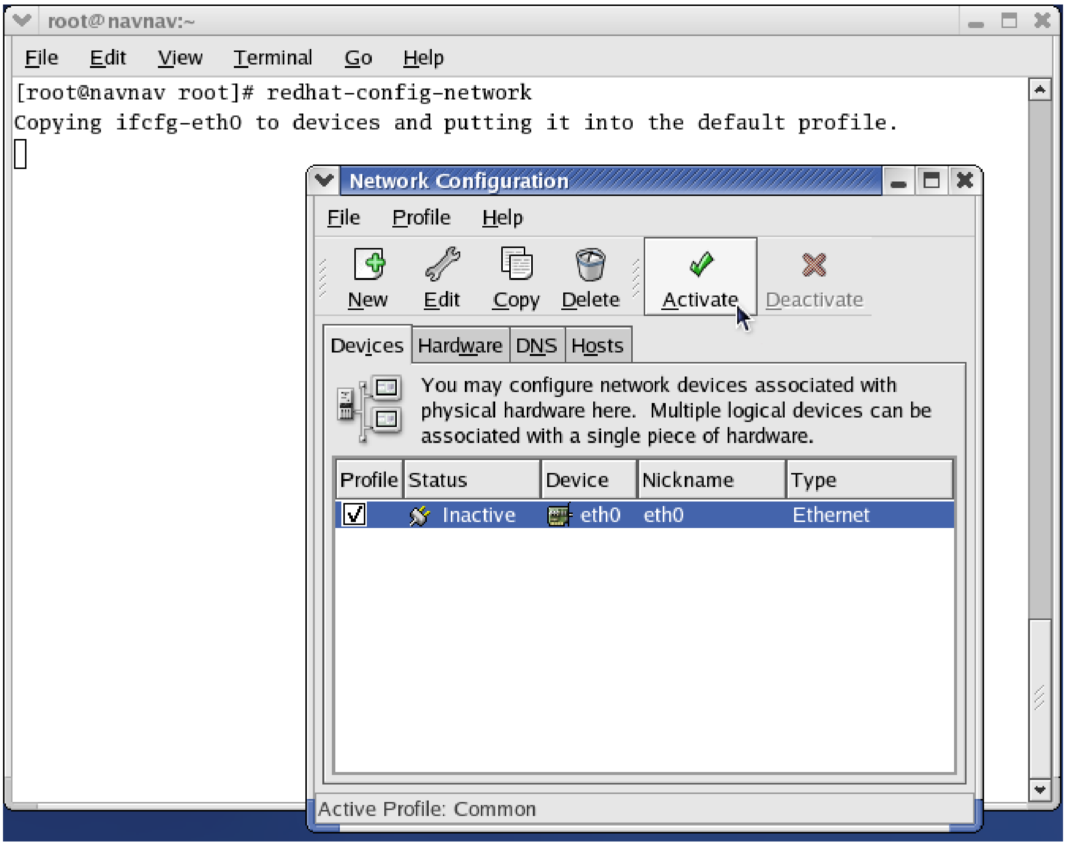



Next, we will open the third file: #gedit/etc/sysconfig/networking/profiles/default/ifcfg-eth0




After modification of these three files, we ned to activate the internet.




Once it is activated, we can browse the internet through web browser.

## 3.3 Domain Name System

- Rather than remember the IP address of the Wrox web site, it is easier for us to remember www.wrox.com.
- Domain Name System (DNS) servers provide the mapping between human−readable addresses (such as www.wrox.com) and the IP addresses of the machines acting as the web servers for the corresponding web service.
- Applications such as web browsers and e−mail clients require the IP address to connect to a web site or a mail server respectively.
- In order to get this from the human−readable input that we provide them with, they query a DNS server for the corresponding IP address information.
- Obviously, this also means that the browser and other clients on the machine need to know the IP address of the DNS server.
- For machines that use DHCP, the information about the DNS server is automatically available when the machine is configured.

Some useful Commands for connecting to internet:

- #redhat-config-network //to open network configuration manager
- ifconfig // to get information about active network
- ping //used to test the reachability of a host

## Keywords

- Network Interfacing Tool is also known as the Network Administration Tool or Network Configuration Tool.
- A network device connected to a TCP/IP network has an IP address associated with it, such as 192.168.100.20. Using the IP address of a machine, other machines on the network can address it uniquely.
- Dynamic IP addresses are distributed and managed via dynamic address allocation protocols.
- Applications such as web browsers and e−mail clients require the IP address to connect to a web site or a mail server respectively.
- For machines that use DHCP, the information about the DNS server is automatically available when the machine is configured.

## Summary

- **Static address**: These are allotted to machines indefinitely and do not change. Typically, static addresses are allocated to servers.
- **Dynamic address**: These are allotted to machines for a specific period with no guarantee that the same address will be available next time the machine connects to the network.
- **Devices Tab**: This tab lists the device connections that we have available on our machine.
- **Hardware Tab**: It allows us to manage the various network devices on the system, such as Ethernet cards, internal modems, and wireless cards.
- **DNS Tab**: It allows us to specify DNS server information.
- **Hosts Tab**: It allows us to modify the hostname of the machine and add aliases to the same host.
- **Domain Name System (DNS)**: It provide the mapping between human−readable addresses (such as www.wrox.com) and the IP addresses of the machines acting as the web servers for the corresponding web service.

## Self Assessment

1. If we want to make a connection to the LAN, then what kind of device will be chosen?

A. CIPC Connection
B. Ethernet Connection
C. ISDN Connection
D. None of the above

2. What is an internet?

A. A tool to write the text data
B. A network that connects the computer all over the world
C. A tool to convert the word doc to pdf
D. None of the above

3. The IP address can be assigned

A. Statically
B. Dynamically
C. Both statically and dynamically

D. None of the above

4. What is another name of Network Interfacing Tool?
A. Network Administration Tool
B. Network Configuration Tool
C. Both network administration and configuration tool
D. None of the above

5. For a machine connected to a LAN or ISP using a static IP address, we need to obtain the network details like
A. IP address
B. Subnet mask
C. Default gateway address
D. All IP address, subnet mask and default gateway address

6. Which of these tabs are available in Network Interfacing Tool?
A. Device tab
B. Hardware tab
C. DNS tab
D. All device, hardware, and DNS tabs

7. In the case of a machine connected to a LAN using a dynamic IP address, the address is allocated either using ____
A. DHCP
B. BOOTP
C. Either DHCP or BOOTP
D. None of the above

8. In the process of activating the internet, how many files were modified?
A. One
B. Two
C. Three
D. Four

9. Which of these is not a web browser?
A. Windows
B. Google Chrome
C. Mozilla Firefox
D. Microsoft Edge

10. How to open the network configuration manager through command line?
A. redhat-config-network
B. redhat-config-internet

C. redhat-config-mozillabrowser

D. None of the above

11. What is used by browsers to retrieve any published resource on the web?

A. URL

B. VRL

C. LRU

D. None of the above

12. What is the path to enter the Network Administration Wizard?

A. Main Menu | System Tools | Internet Configuration Wizard

B. Main Menu | Systems | Settings

C. Main Menu | System Tools | Web Browsers

D. None of the above

13. What is the format of IP address?

A. x.x

B. x.x.x

C. x.x.x.x

D. None of the above

14. What is required to access internet?

A. Pdf converter

B. Photoshop

C. Web browser

D. None of the above

15. What provides the interfacing between human-readable address and IP address of the machine?

A. DNS

B. ABC

C. Wrox

D. None of the above

## Answers for Self Assessment

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | C | 2. | B | 3. | C | 4. | C | 5. | D |
| 6. | D | 7. | C | 8. | C | 9. | A | 10. | A |
| 11. | A | 12. | A | 13. | C | 14. | C | 15. | A |

## Review Questions:

1. Is it necessary to configure the system before the internet connection take place? If yes, how can we configure it?
2. What is an IP address? Explain the difference between static and dynamic IP address.
3. What is a network interfacing tool? Explain its tabs.
4. What is DNS? Explain.
5. Explain the process of configuring the system for internet connection.

## Further Readings

Mark G. Sobell, A Practical Guide to Fedora and Red Hat Enterprise Linux, Fifth Edition, Pearson Education, Inc.

**Web Links**

https://www.redhat.com/sysadmin/network-interface-linux